Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Borges Dairy LLC (former John & Teresa Borges Dairy)

Physical address of dairy:

Number and Street	City	County	Zip Code
25834 Kasson RD	Tracy	San Joaquin	95304

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1938

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X239-X150-XX01-XXXX

B. OPERATORS

Borges, Carlos

Operator name: Borges, Carlos

Telephone no.: Landline: ; Cellular: (209) 652-1126

Mailing Address Number and Street	City	State	Zip Code
25822 Kasson RD	Tracy	CA	95304

C. OWNERS

Borges, John

Legal owner name: Borges, John

Telephone no.: Landline: (209) 835-6160; Cellular:

Mailing Address Number and Street	City	State	Zip Code
25834 Kasson RD	Tracy	CA	95304

This owner is responsible for paying permit fees.

Borges, Teresa

Legal owner name: Borges, Teresa

Telephone no.: Landline: (209) 835-6160; Cellular:

Mailing Address Number and Street	City	State	Zip Code
25834 Kasson RD	Tracy	CA	95304

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	58	115	120	50	45
Number under roof	400	0	0	0	0	0
Maximum number	420	60	120	125	60	50
Average number	400	58	115	120	50	45
Avg live weight (lbs)	1,400	1,450	1,400	900		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,769.54 *tons per reporting period*

Total nitrogen from manure: 168,824.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 118,176.88 *lbs per reporting period*

Total phosphorus from manure: 27,661.45 *lbs per reporting period*

Total potassium from manure: 75,289.69 *lbs per reporting period*

Total salt from manure: 201,677.10 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,886,764 *gallons*

Total nitrogen generated: 56,867.06 *lbs*

Total phosphorus generated: 7,329.61 *lbs*

Total potassium generated: 100,291.00 *lbs*

Total salt generated: 521,502.50 *lbs*

	9,886,764	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,886,764	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	39	39	2	both	X239-X150-XX01-XXXX
Field 2	38	38	2	both	X239-X150-XX01-XXXX
Field 3	55	55	2	both	X239-X150-XX01-XXXX
Field 4	75	75	2	both	X239-X150-XX01-XXXX
Totals for areas that were used for application	207	207	8		
Totals for areas that were not used for application					
Land application area totals	207	207	8		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	349.98 *ton*	As-is		61.0	8,130.00	1,260.00	7,690.00		9.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.61	172.44	26.73	163.11	813.99

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	790.58 *ton*	As-is		73.4	2,930.00	500.00	3,720.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	23.96	140.39	23.96	178.24	904.90

Field 2

Field name: Field 2

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	472.28 *ton*	As-is		67.0	4,480.00	1,060.00	6,440.00		9.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.43	111.36	26.35	160.08	800.59

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	906.41 *ton*	As-is		69.8	3,260.00	620.00	4,530.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	23.85	155.52	29.58	216.11	907.65

Field 3

Field name: Field 3

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	545.60 *ton*	As-is		63.2	5,660.00	1,370.00	8,180.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	9.92	112.29	27.18	162.29	759.32

Field 3

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	891.11 *ton*	As-is		69.2	3,820.00	590.00	3,960.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	16.20	123.78	19.12	128.32	638.75

Field 4

Field name: Field 4

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	678.80 *ton*	As-is		60.9	6,120.00	1,350.00	8,300.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	9.05	110.78	24.44	150.24	750.23

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,646.00 *ton*	As-is		69.2	4,130.00	640.00	3,110.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.95	181.28	28.09	136.51	865.23

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/17/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.85	10.08	90.00	752.02	406,262.00 *gal*
Application event totals		46.85	10.08	90.00	752.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	127.12	22.67	107.20	574.88	375,136.00 *gal*
Application event totals		127.12	22.67	107.20	574.88	

Field 1 - 06/15/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	76.25	26.03	164.99	2,925.13	220.00 *ton*
Application event totals		76.25	26.03	164.99	2,925.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.76	3.38	147.71	644.36	423,269.00 *gal*
Canal	Surface water	0.00	0.00	0.00	156.61	9,527,755.00 *gal*
Application event totals		58.76	3.38	147.71	800.96	

Field 1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	65.43	3,980,753.00 *gal*
Application event totals		0.00	0.00	0.00	65.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.38	3.30	144.24	629.22	413,326.00 *gal*
Canal	Surface water	0.00	0.00	0.00	62.15	3,780,807.00 *gal*
Application event totals		57.38	3.30	144.24	691.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	62.15	3,780,807.00 *gal*
Application event totals		0.00	0.00	0.00	62.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	61.22	3,724,761.00 *gal*
Application event totals		0.00	0.00	0.00	61.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	40.33	2,453,889.00 *gal*
Application event totals		0.00	0.00	0.00	40.33	

Field 2 - 11/17/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.32	9.32	83.22	695.40	432,592.00 *gal*
Application event totals		43.32	9.32	83.22	695.40	

01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.18	19.12	90.39	484.72	364,232.00 *gal*
Application event totals		107.18	19.12	90.39	484.72	

Field 2 - 06/15/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	75.25	25.69	162.82	2,886.64	250.00 *ton*
Application event totals		75.25	25.69	162.82	2,886.64	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.76	3.56	155.25	677.27	512,296.00 *gal*
Canal	Surface water	0.00	0.00	0.00	154.93	10,853,522.00 *gal*
Application event totals		61.76	3.56	155.25	832.19	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	66.49	4,658,268.00 *gal*
Application event totals		0.00	0.00	0.00	66.49	

Field 2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.23	3.47	151.39	660.43	499,562.00 *gal*
Canal	Surface water	0.00	0.00	0.00	63.38	4,440,110.00 *gal*
Application event totals		60.23	3.47	151.39	723.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	60.95	4,269,596.00 *gal*
Application event totals		0.00	0.00	0.00	60.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	63.41	4,441,972.00 *gal*
Application event totals		0.00	0.00	0.00	63.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	43.32	3,034,528.00 *gal*
Application event totals		0.00	0.00	0.00	43.32	

Field 3 - 11/18/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.67	8.96	80.05	668.93	602,293.00 *gal*
Application event totals		41.67	8.96	80.05	668.93	

Field 3 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.12	20.00	94.55	507.05	551,462.00 *gal*
Application event totals		112.12	20.00	94.55	507.05	

Field 3 - 06/19/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	74.86	25.56	161.99	2,871.95	360.00 *ton*
Application event totals		74.86	25.56	161.99	2,871.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.72	3.73	162.68	709.67	776,959.00 *gal*
Canal	Surface water	0.00	0.00	0.00	70.13	7,111,138.00 *gal*
Application event totals		64.72	3.73	162.68	779.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	69.50	7,046,566.00 *gal*
Application event totals		0.00	0.00	0.00	69.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	73.30	7,432,149.00 *gal*
Application event totals		0.00	0.00	0.00	73.30	

Field 3 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.53	2.91	127.02	554.09	606,629.00 *gal*
Canal	Surface water	0.00	0.00	0.00	73.30	7,432,149.00 *gal*
Application event totals		50.53	2.91	127.02	627.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	61.60	6,246,193.00 *gal*
Application event totals		0.00	0.00	0.00	61.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	60.78	6,162,945.00 *gal*
Application event totals		0.00	0.00	0.00	60.78	

Field 4 - 11/18/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.83	8.57	76.51	639.30	784,929.00 *gal*
Application event totals		39.83	8.57	76.51	639.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.94	19.25	91.02	488.15	723,959.00 *gal*
Application event totals		107.94	19.25	91.02	488.15	

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	74.73	25.51	161.69	2,866.63	490.00 *ton*
Application event totals		74.73	25.51	161.69	2,866.63	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.21	4.79	209.16	912.41	1,362,162.00 *gal*
Canal	Surface water	0.00	0.00	0.00	81.84	11,315,248.00 *gal*
Application event totals		83.21	4.79	209.16	994.25	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	87.28	12,068,077.00 *gal*
Application event totals		0.00	0.00	0.00	87.28	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	71.56	9,894,654.00 *gal*
Application event totals		0.00	0.00	0.00	71.56	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.24	3.70	161.49	704.45	1,051,696.00 *gal*
Canal	Surface water	0.00	0.00	0.00	61.21	8,462,696.00 *gal*
Application event totals		64.24	3.70	161.49	765.66	

Field 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	53.90	7,452,196.00 *gal*
Application event totals		0.00	0.00	0.00	53.90	

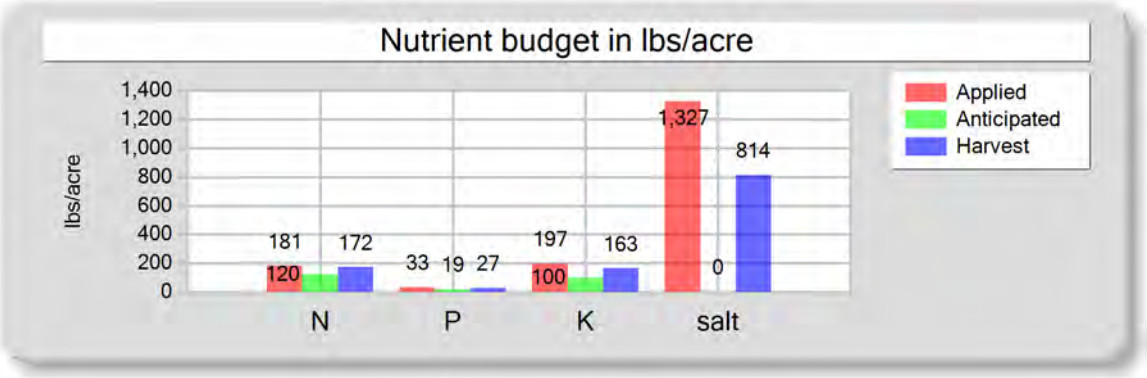
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	44.56	6,161,139.00 *gal*
Application event totals		0.00	0.00	0.00	44.56	

B. NUTRIENT BUDGET

Field 1 - 11/17/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.96	32.75	197.19	1,326.90
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.96	32.75	197.19	1,326.90
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	172.44	26.73	163.11	813.99
Nutrient balance	8.52	6.03	34.08	512.91
Applied to removed ratio	1.05	1.23	1.21	1.63

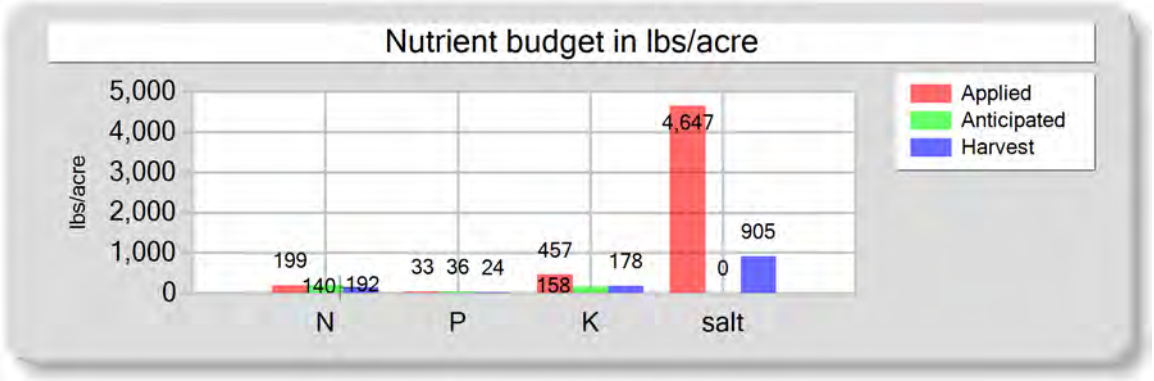
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
781,398.00	*gallons*
28.78	*acre-inches*
0.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/15/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	76.25	26.03	164.99	2,925.13
Process wastewater	116.14	6.69	291.95	1,273.57
Fresh water	0.00	0.00	0.00	447.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.40	32.72	456.94	4,646.60
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	140.39	23.96	178.24	904.90
Nutrient balance	59.01	8.76	278.70	3,741.70
Applied to removed ratio	1.42	1.37	2.56	5.13

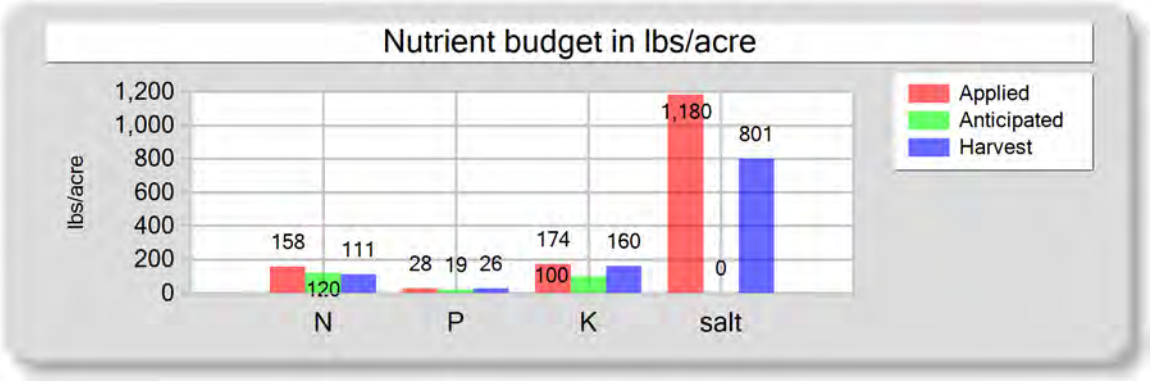
Fresh water applied	
27,248,772.00	*gallons*
1,003.48	*acre-inches*
30.41	*inches/acre*

Process wastewater applied	
836,595.00	*gallons*
30.81	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/17/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.50	28.44	173.61	1,180.12
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.50	28.44	173.61	1,180.12
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	111.36	26.35	160.08	800.59
Nutrient balance	46.14	2.09	13.53	379.53
Applied to removed ratio	1.41	1.08	1.08	1.47

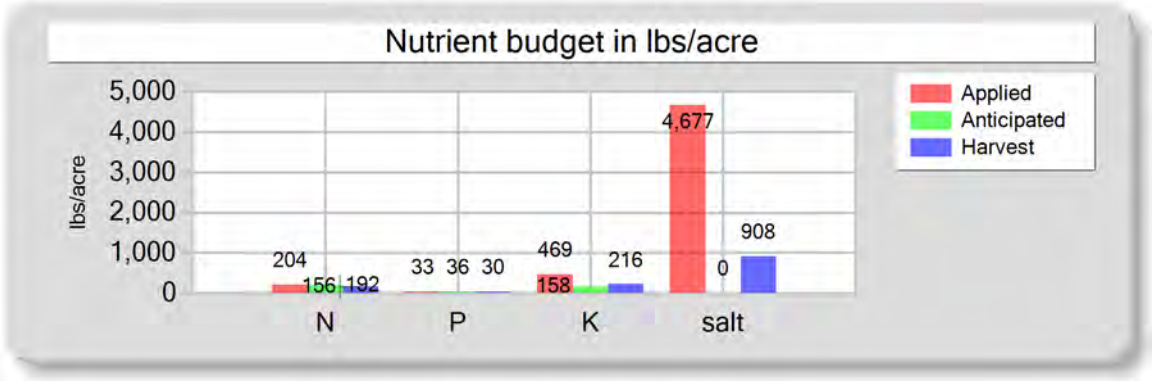
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
796,824.00	*gallons*
29.34	*acre-inches*
0.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/15/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	75.25	25.69	162.82	2,886.64
Process wastewater	121.99	7.02	306.65	1,337.70
Fresh water	0.00	0.00	0.00	452.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	204.24	32.71	469.47	4,676.81
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	155.52	29.58	216.11	907.65
Nutrient balance	48.72	3.13	253.36	3,769.16
Applied to removed ratio	1.31	1.11	2.17	5.15

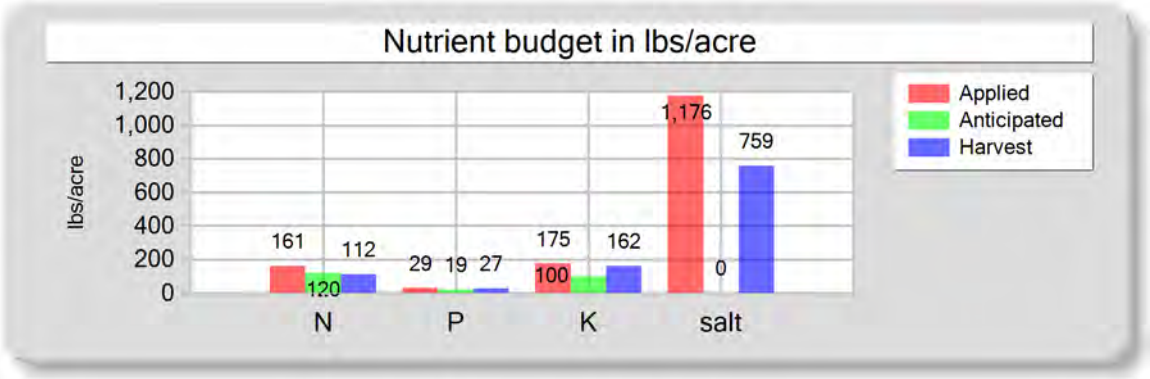
Fresh water applied	
31,697,996.00	*gallons*
1,167.33	*acre-inches*
30.72	*inches/acre*

Process wastewater applied	
1,011,858.00	*gallons*
37.26	*acre-inches*
0.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/18/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.79	28.96	174.60	1,175.98
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	160.79	28.96	174.60	1,175.98
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	112.29	27.18	162.29	759.32
Nutrient balance	48.50	1.78	12.31	416.67
Applied to removed ratio	1.43	1.07	1.08	1.55

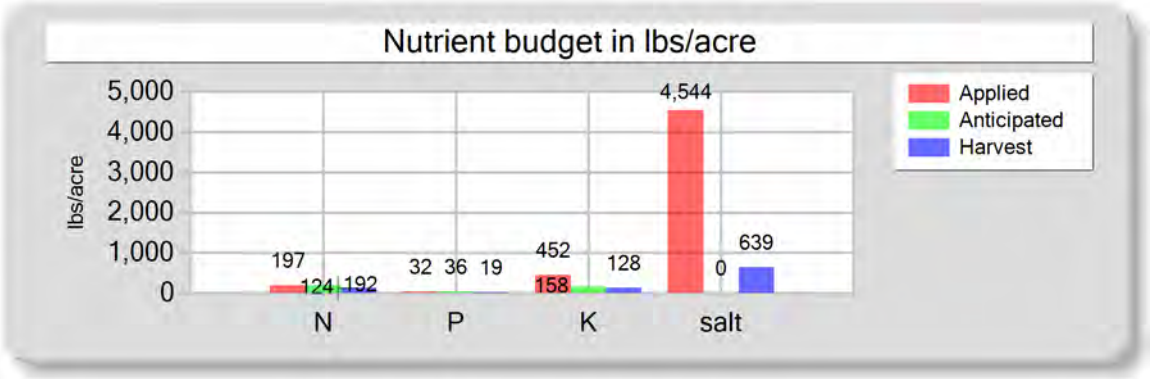
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,153,755.00	*gallons*
42.49	*acre-inches*
0.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/19/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	74.86	25.56	161.99	2,871.95
Process wastewater	115.25	6.63	289.70	1,263.77
Fresh water	0.00	0.00	0.00	408.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.11	32.19	451.69	4,544.32
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	123.78	19.12	128.32	638.75
Nutrient balance	73.33	13.07	323.37	3,905.57
Applied to removed ratio	1.59	1.68	3.52	7.11

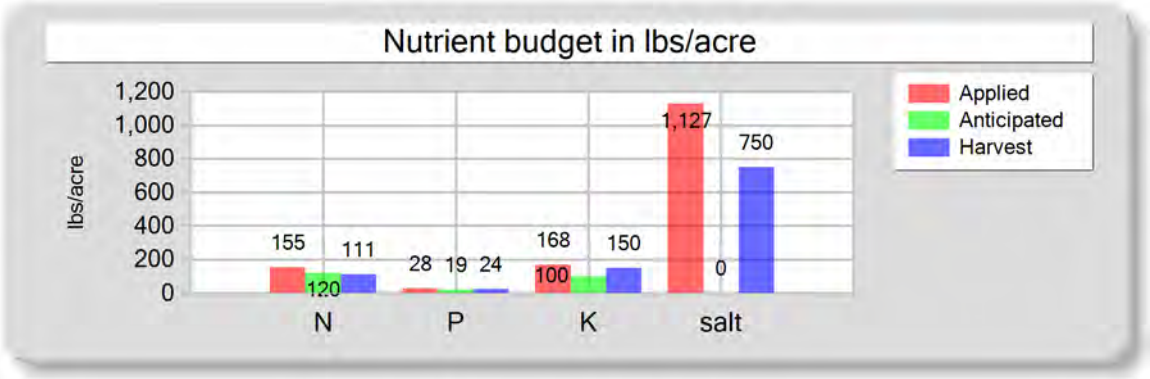
Fresh water applied	
41,431,140.00	*gallons*
1,525.77	*acre-inches*
27.74	*inches/acre*

Process wastewater applied	
1,383,588.00	*gallons*
50.95	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/18/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	147.77	27.82	167.53	1,127.45
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.77	27.82	167.53	1,127.45
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	110.78	24.44	150.24	750.23
Nutrient balance	43.99	3.38	17.29	377.22
Applied to removed ratio	1.40	1.14	1.12	1.50

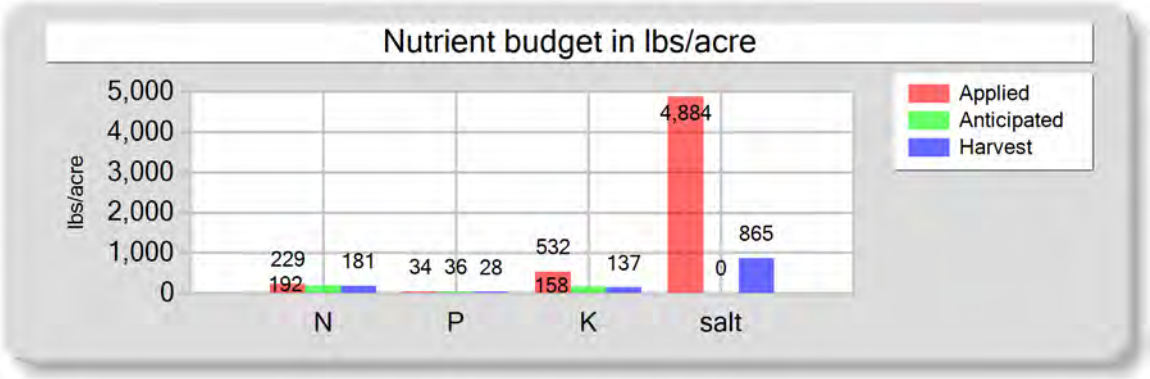
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,508,888.00	*gallons*
55.57	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/20/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	74.73	25.51	161.69	2,866.63
Process wastewater	147.45	8.49	370.64	1,616.86
Fresh water	0.00	0.00	0.00	400.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.18	34.00	532.33	4,883.83
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	181.28	28.09	136.51	865.23
Nutrient balance	47.90	5.91	395.83	4,018.61
Applied to removed ratio	1.26	1.21	3.90	5.64

Fresh water applied	
55,354,010.00	*gallons*
2,038.50	*acre-inches*
27.18	*inches/acre*

Process wastewater applied	
2,413,858.00	*gallons*
88.89	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 50.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	3,960.00	25,100.00	0.00	0.00	0.00	0.00	0.00		44.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 25.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,000.00	5,810.00	30,600.00	0.00	0.00	0.00	0.00	0.00		48.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4
Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	302.00	0.00	0.00	98.10	876.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,540.00	7,320
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 01/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,340.00	456.00	0.00	0.00	239.00	1,130.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9,590.00	6,060
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,730.00	546.00	0.00	0.00	169.00	1,250.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11,800.00	7,580
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	417.00	0.00	0.00	31.60	1,380.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11,500.00	6,020
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	490.00	356.00	0.00	0.00	74.60	702.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,920.00	5,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 08/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	217.00	65
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,130.00	1,260.00	7,690.00		9.84
DL	100.00	100.00	100.00		0.01

Field 1 - 06/15/2023: Corn, silage

Field 1 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,930.00	500.00	3,720.00		7.10
DL	100.00	100.00	100.00		0.01

Field 2 - 11/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	1,060.00	6,440.00		9.76
DL	100.00	100.00	100.00		0.01

Field 2 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	620.00	4,530.00		6.30
DL	100.00	100.00	100.00		0.01

Field 3 - 11/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,660.00	1,370.00	8,180.00		10.40
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,580.00	1,410.00	8,610.00		10.20
DL	100.00	100.00	100.00		0.01

Field 3 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,280.00	600.00	3,720.00		6.30
DL	100.00	100.00	100.00		0.01

Field 3 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,820.00	590.00	3,960.00		6.40
DL	100.00	100.00	100.00		0.01

Field 4 - 11/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,120.00	1,350.00	8,300.00		10.60
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,880.00	1,360.00	7,890.00		9.79
DL	100.00	100.00	100.00		0.01

Field 4 - 06/20/2023: Corn, silage

Field 4 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,130.00	640.00	3,110.00		6.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,090.00	500.00	4,410.00		6.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

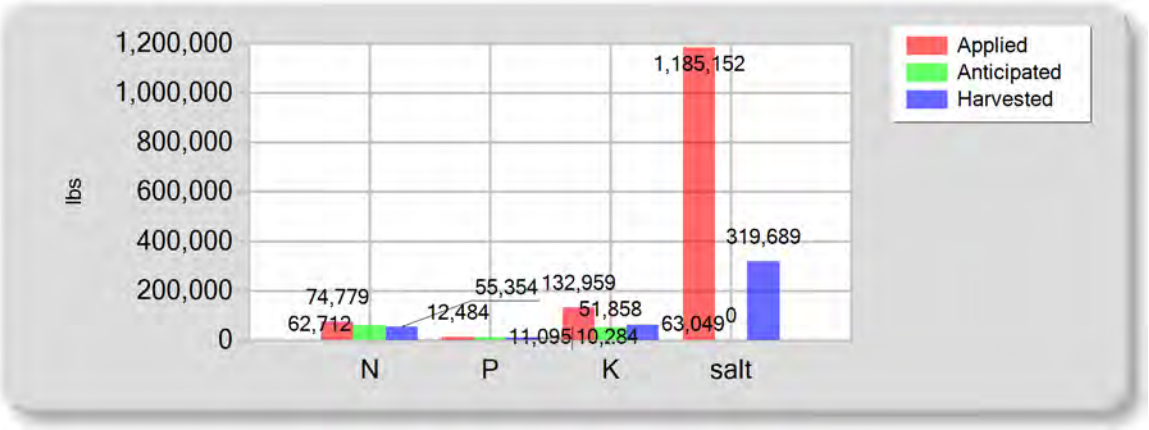
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

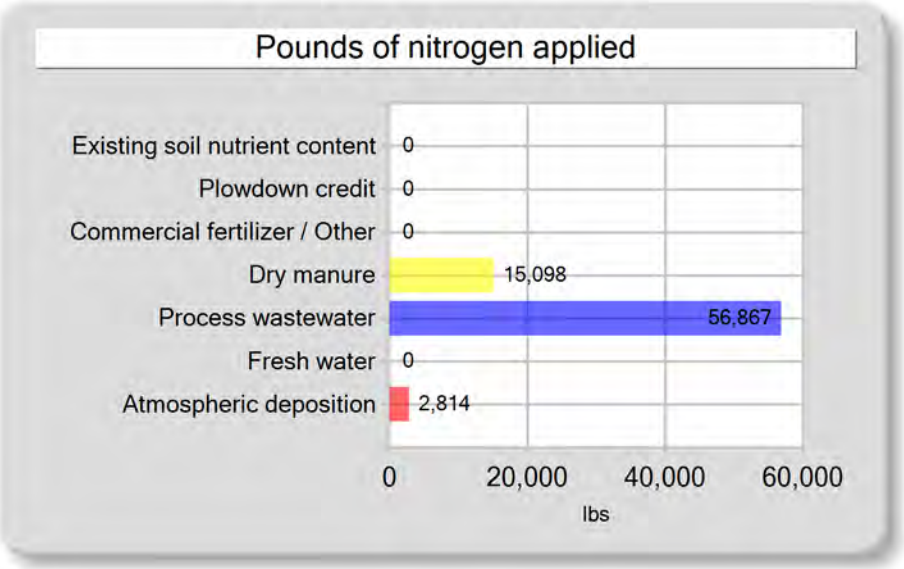
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

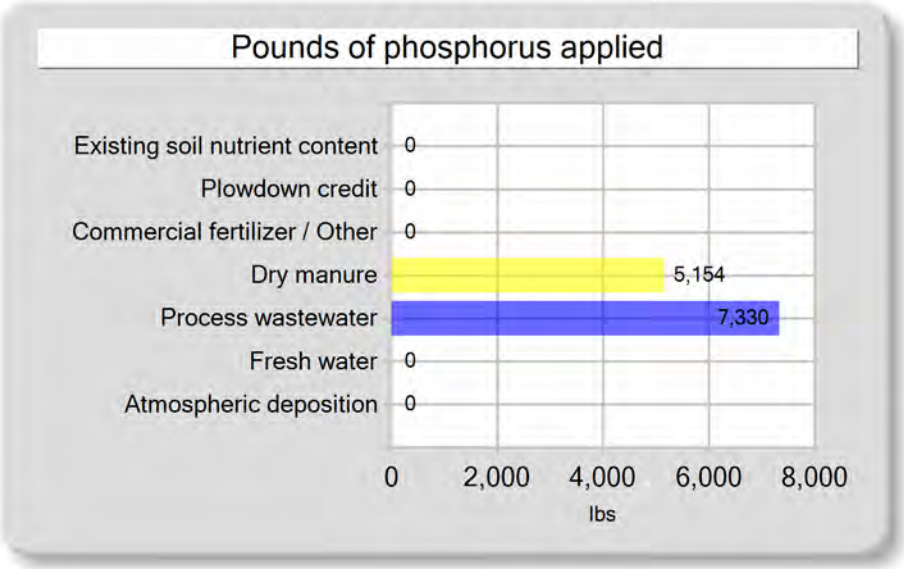
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,097.63	5,154.02	32,668.15	579,176.40
Process wastewater	56,867.06	7,329.61	100,291.00	521,502.50
Fresh water	0.00	0.00	0.00	84,472.89
Atmospheric deposition	2,814.00	0.00	0.00	0.00
Total nutrients applied	74,778.69	12,483.62	132,959.15	1,185,151.78
Anticipated crop nutrient removal	62,712.00	11,095.20	51,858.00	0.00
Actual crop nutrient removal	55,353.64	10,283.81	63,049.46	319,689.08
Nutrient balance	19,425.05	2,199.82	69,909.70	865,462.71
Applied to removed ratio	1.35	1.21	2.11	3.71

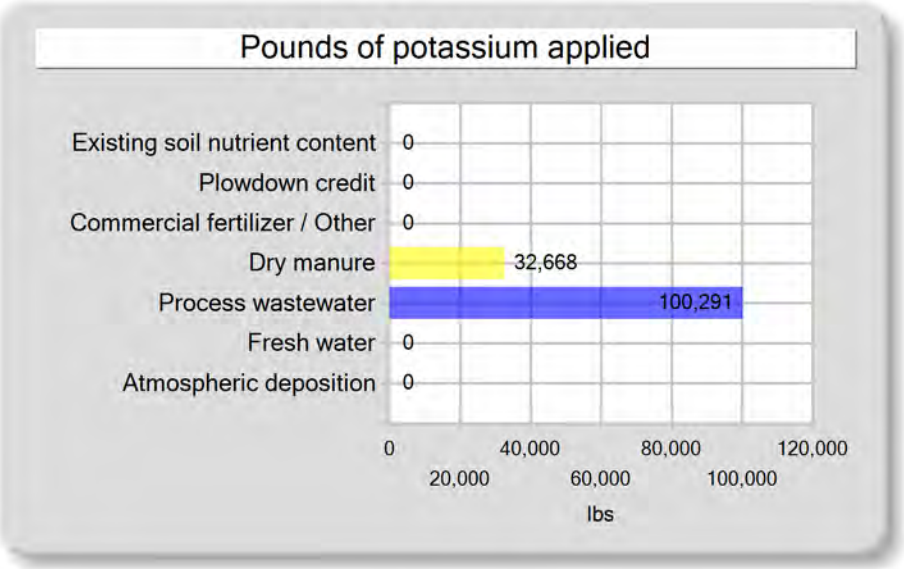
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

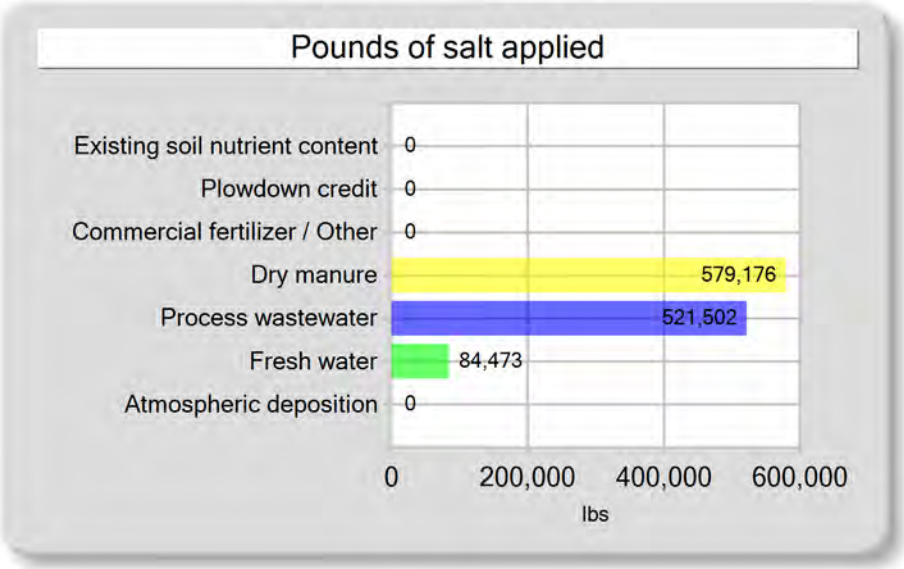


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

A high ratio was recorded on Field 3 for the Corn Silage crop. This was due to an under performing yield of 16.2 tons per acre compared to the anticipated 24 Tons per acre.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Borges	Carlos Borges (Apr 23, 2024 09:04 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Borges	Carlos Borges
PRINT OR TYPE NAME	PRINT OR TYPE NAME
4-18-24	Apr 23, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
John Borges Dairy
25822 Kasson Rd
Tracy, CA 95304

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Borges Dairy LLC
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Milk Barn	T3079004A	Electrical Conductivity (EC)	EPA 120.1 and	3.29	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2100	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Compliance Analysis Report

Customer
John Borges Dairy
25822 Kasson Rd
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Grower

Consultant:
F & R Ag Services
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PURCHASE ORDER: Borges Dairy LLC
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-#2	T3079004B	Electrical Conductivity (EC)	EPA 120.1 and	3.53	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	13.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2260	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: John BORGES DAIRY LLC
Mailing address:
(not facility address) TRACY CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/17/23	9:40	DW - MILK BARN	DOMESTIC	3.4°C		X				
2	"	9:45	DW - #2	"	3.2°C		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-20-23 7:30			T3079004

Required to be submitted to any agency? (YES / NO)

2023 Borges Dairy Annual Report

Final Audit Report 2024-04-23

Created:	2024-04-22
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAkLaAVyOacEbQTy0Nyvnb7i1jUowrJc6A

"2023 Borges Dairy Annual Report" History

Document created by Sean Locke (slocke@fragservices.com)
2024-04-22 - 7:35:34 PM GMT

Document emailed to borgesdairy@outlook.com for signature
2024-04-22 - 7:37:06 PM GMT

Email viewed by borgesdairy@outlook.com
2024-04-22 - 8:18:10 PM GMT

Signer borgesdairy@outlook.com entered name at signing as Carlos Borges
2024-04-23 - 4:04:39 PM GMT

Document e-signed by Carlos Borges (borgesdairy@outlook.com)
Signature Date: 2024-04-23 - 4:04:41 PM GMT - Time Source: server

Agreement completed.
2024-04-23 - 4:04:41 PM GMT